

Eduardo Arias | CV

Jipijapa – Quito – Ecuador

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Professional Summary

Mathematical Engineer and data scientist. My work closes two gaps that are usually open: between statistical rigor and production code, and between experimental AI and auditable AI. At CEIE (Center for Evaluation and Educational Innovation, Universidad UTE) I am responsible for all quantitative analytics —data preparation, pipelines, psychometric and ML modeling— that feeds institutional decision-making: ~20 repositories in production, ETL, IRT, causal inference, BI, and local LLM/RAG systems. I work by one rule: nothing reaches production without formal evidence justifying the method and tests guaranteeing the result — an IRT model has regression tests, a Bayesian decision documents its priors, a local LLM leaves per-call traceability. Creator of Aurora-GLM (GLMs with automated numerical validation against R and statsmodels) and author of a publication on nonlinear PDEs.

Experience

CEIE, Universidad UTE

Quantitative Analyst / Data Scientist

Quito

2024–present

Responsible for all quantitative analytics at the Center for Evaluation and Educational Innovation —data preparation, pipelines, psychometric and ML modeling— whose output feeds university leadership decisions: surveys of 15,000+ students, ~1,300 faculty per term, and 23 degree programs across 5 assessment cycles.

- Designed and maintain an **ecosystem of ~20 production repositories** covering the full cycle: capture → ETL → psychometric analysis → BI → mass reporting, with a Medallion architecture (21 Parquet tables in star schema) and 94 exact-regression golden tests.
- **IRT equating** (2PL/3PL, mirt/TAM) with anchor items across exam forms; **BIRT** (brms) with empirical priors; CFA with competing models (lavaan) and causal analysis (PSM, AIPW doubly-robust) over 4,965 students × 214 variables.
- Predictive models for **student dropout** (LightGBM/XGBoost, SMOTE, Cox/Kaplan–Meier survival) and a 3-stage NLP cascade (VADER → spaCy → Transformers) over thousands of open-ended comments.
- **8 Power BI semantic models** (PBIP/TMDL) with native psychometric metrics in DAX and multilevel percentile benchmarks centralizing institutional analytics.
- **Mass reporting**: 1,000 individualized PDFs in ~13 min (99% success) with SMTP delivery and per-recipient tracking.
- **100% local AI**: empirical validation of 6 RAG architectures (LlamaIndex, Qdrant, pgvector, BGE-M3) over institutional corpora; automatic curriculum maps (DAG graphs) of 23 degree programs with embeddings + LLM-as-a-judge + offline Wikipedia (6.09M articles).
- **Institutional integration and data quality**: proprietary connectors for Microsoft Graph API (SharePoint/Forms/OneDrive), Moodle, Banner, SICAF, and KOBO; ~42 automated audit validations per run with quality reports and documented gap handling.
- **Reproducible research**: causal pipeline of ~100 scripts in 7 phases with quality gates and MLflow tracking; Bayesian ranking (WAIC) of 8 RIASEC index operationalizations; author of institutional methodological guides and a technical book on TDA/IRT (8 chapters) in $\text{\LaTeX}$.
- Full-stack platform **CEIE-Evaluaciones** (FastAPI, Vue 3, dual PostgreSQL + MongoDB with bidirectional sync, Celery/Redis, Docker Compose with 8+ services).

Willay

Data Scientist

2023–2024

- Founded the ML area: designed and built from scratch the AI backend and an insurability index for vehicle insurers in production, integrating geospatial (H3), demographic, and risk variables with probabilistic models (Pyro).
- Led a team of 2 analysts in backend development, testing, and data exploration; designed AI models for strategic risk-assessment decisions.
- Administered AWS infrastructure for the ML server; developed a scalable backend for data pipelines.

Grupo Saroquis

Data Analyst

Lumbisí

2022–2023

- Machine Learning and Deep Learning models for algorithmic trading: comprehensive data analysis, synthetic indices, and predictive modeling.
- Developed and implemented fully automated buy/sell flows with average returns of 65–77% in fully automated operations.
- Automated pipelines for execution, monitoring, and optimization of algorithmic strategies.

Projects

Aurora-GLM: Python framework for GLM, GAM, and GAMM published on PyPI (v1.0.0), designed and developed as an individual research project. **Automated numerical validation against R and statsmodels in CI:** maximum deviation 1.4×10^{-13} across 20 family×replication comparisons; 3,379-test suite, clean mypy/ruff. Multi-backend API (NumPy, PyTorch, JAX), 9 distribution families, R-style formulas; GPU speedup of 4.5–5.9× at n=50,000 verified on RTX 5070 Ti (an earlier 141× claim from older hardware did not reproduce and is documented as such). MIT license. GitHub.

Education and Certifications

Florida Global University / GEM <i>International Specialization in Data Science for Banking and Finance</i> 320 academic hours.	2026
BEPS Smart Research (Microsoft Partner), Lima <i>International Diploma in Data Engineering</i> 720 academic hours. Modules: Data Storage Essentials, Artificial Intelligence with Python, Engineering for Massive Data Processing, Massive Data Visualization, Web Scraping, Cloud Computing with Azure.	2026
BEPS Smart Research, Lima <i>Linear Regression with Time Series</i> 97 academic hours.	2025
Chilean Statistical Society (SOCHE) <i>Introduction to Spatial Econometrics</i> August 2025.	2025
Ecuadorian Statistical Society <i>Machine Learning with Tree-Based Ensemble Models in R</i> 20 academic hours.	2024
Ecuadorian Data Scientists College (CCDE) <i>Deep Learning with Python</i> 25 academic hours.	2022
Escuela Politécnica Nacional <i>Mathematical Engineering, Cum Laude</i> Modeling and Scientific Computing. SENECYT 1001-2021-2300449.	Quito

Technical Skills

Languages: Python (FastAPI, PyTorch, pandas, scikit-learn), R (mirt, TAM, lavaan, brms, tidyverse), SQL, DAX/Power Query M, JavaScript (Vue 3), $\text{\LaTeX}$.

ML/Statistics: IRT/BIRT and equating, CFA, causal inference (PSM/AIPW), XGBoost/LightGBM, survival analysis, NLP, PINNs, local LLMs and RAG (llama.cpp, BGE-M3).

Infrastructure: PostgreSQL, MongoDB, Redis, Docker, AWS, Microsoft Azure, Power BI (PBIP/TMDL), MLflow, Git.

Languages: Spanish (native), English (B2).

Publication

2024: E. Arias, M. Calahorrano, A. Castro. *A biharmonic equation with discontinuous nonlinearities*. Electron. J. Diff. Eq., Vol. 2024, No. 15. — Existence and multiplicity analysis for higher-order PDEs with discontinuous nonlinearities; demonstrates rigorous analytical reasoning applicable to modeling complex phenomena with discontinuities in data.